Academic Progress Report: ML-Based Predictive Modelling for LMD Process Optimisation

1. Executive Summary

This exercise implemented and rigorously evaluated six machine learning models — GPR, SVR, RF, ANN, Stacking V1 (RF+ANN), and Stacking V2 (GPR+ANN) — for predicting two key bead geometry outputs of the Laser Metal Deposition (LMD) process: **bead height (µm)** and **aspect ratio**. Working with a dataset of **N=89 experimental samples**, the study confirms that **Gaussian Process Regression (GPR) is the dominant model** for both targets, achieving R^2=0.9552 for height and R^2=0.8739 for aspect ratio on out-of-fold cross-validation. Critically, a **Multi-Output GPR** further improved performance to R^2=0.9682 and R^2=0.9689 respectively, providing the most significant finding of this exercise: *bead height and aspect ratio are not independent — they share physics-driven variance that joint modelling can exploit.*

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2. Dataset & Feature Engineering

The dataset consists of **89 samples** with three controllable process inputs — Laser Power (W), Scan Speed (mm/s), and Powder Feed Rate (g/min) — and a **physics-informed derived feature**, Heat Input (P/S) . This engineered feature represents Linear Energy Density (LED), the energy delivered per unit length of the scan track.[ppl-ai-file-upload.s3.amazonaws]

Feature	Description	Type
Power (W)	Laser power	Raw input
Speed (mm/s)	Scan speed	Raw input
Feed (g/min)	Powder feed rate	Raw input
Heat Input (P/S)	Linear Energy Density	Physics-derived

The target range is substantial: height spans **214–2432 µm** (mean 803 ± 426 µm) and aspect ratio spans **0.078–1.402** (mean 0.342 ± 0.215). This wide spread is actually positive — it means the design of experiments (DoE) explored a broad process space, giving models enough variation to learn from. However, it also means prediction at extreme values remains a risk zone.[ppl-ai-file-upload.s3.amazonaws]

All features and targets were **standardised (zero mean, unit variance)** before modelling. This is a mandatory step for GPR, SVR, and ANN, as these models are sensitive to feature scale. Crucially, targets were scaled in a **separate scaler** and back-transformed to original units for all reported metrics, ensuring all RMSE and MAE values are physically interpretable in µm and dimensionless units respectively.[ppl-ai-file-upload.s3.amazonaws]

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3. Core Model Performance Results

Height Prediction (ranked by MAE)

Model	R^2 (OOF)	R^2 (CV)	CV–OOF Gap	RMSE (µm)	MAE (µm)	MAPE (%)
GPR	**0.9552**	**0.9437**	**0.012**	**90.2**	**64.6**	**9.53**
Stack V2 (GPR+ANN)	0.9449	0.9300	0.015	100.1	71.9	10.30
ANN	0.9401	0.9184	0.022	104.3	78.3	11.25
Stack V1 (RF+ANN)	0.9048	0.8850	0.020	131.5	94.3	13.22
SVR	0.7509	0.8300	0.079	212.7	110.7	15.43
RF	0.6844	0.7561	0.072	239.4	132.4	17.80

Aspect Ratio Prediction (ranked by MAE)

Model	R^2 (OOF)	R^2 (CV)	CV–OOF Gap	RMSE	MAE	MAPE (%)
GPR	**0.8739**	**0.9083**	**0.034**	**0.0765**	**0.0379**	**10.79**
ANN	0.8039	0.8787	0.075	0.0953	0.0416	10.10
Stack V2 (GPR+ANN)	0.7740	0.8405	0.067	0.1024	0.0489	13.82
SVR	0.5815	0.7910	**0.209**	0.1393	0.0566	14.47
Stack V1 (RF+ANN)	0.6596	0.7898	**0.130**	0.1256	0.0583	15.83
RF	0.5273	0.7253	**0.198**	0.1481	0.0635	17.54

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4. Critical Observations & Key Learnings

GPR's Dominance is Theoretically Well-Founded

GPR is a **Bayesian non-parametric method** that places a prior distribution over functions and updates it with observed data. At small N (here N=89), it outperforms tree-based and neural methods because it does not waste model capacity on spurious patterns — the RBF kernel imposes a smooth inductive bias directly aligned with the smooth physical relationships in LMD (process parameters vary continuously, and so do bead geometry responses). GPR's CV–OOF gap of just **0.012 for height** is the lowest of all models, confirming it generalises reliably and is not overfitting. [ppl-ai-file-upload.s3.amazonaws]

Random Forest Fails at N=89 — A Critical Lesson

RF achieved R^2=0.6844 for height and R^2=0.5273 for aspect ratio — the weakest performance of all models. This is a critical finding: **tree-based ensemble methods require large N to benefit from variance reduction through bagging**. At N=89, each of the 5 CV folds trains on only ~71 samples. Decision trees are high-variance learners; at this sample size, individual trees overfit their training splits and the ensemble cannot average out the instability. The large CV–OOF gap (0.072 for height, 0.198 for aspect ratio) further confirms this overfitting behaviour. This is an important insight for the thesis — it validates the choice of probabilistic models (GPR) over tree methods for small experimental datasets in materials science.[ppl-ai-file-upload.s3.amazonaws]

The SVR and RF Overfitting Red Flag on Aspect Ratio

Both SVR and RF show **extremely high CV–OOF gaps on aspect ratio** (0.209 and 0.198 respectively). This means their cross-validation R^2 (~0.79 and ~0.73) is misleadingly optimistic relative to their actual generalisation performance (~0.58 and ~0.53). This phenomenon — where CV overestimates real-world performance — occurs when a model is too flexible relative to the size of the training data. For the thesis, this reinforces the importance of always reporting both the CV score *and* the out-of-fold score: **reporting CV R^2 alone would overstate SVR and RF performance by ~0.20 points**.[ppl-ai-file-upload.s3.amazonaws]

Stacking V2 (GPR+ANN) vs. V1 (RF+ANN) — Ensemble Design Matters

Replacing RF with GPR in the stacking ensemble yielded a meaningful improvement: on height, Stack V2 achieves R^2=0.9449 vs Stack V1's 0.9048 (Δ=+0.040); on aspect ratio, 0.7740 vs 0.6596 (Δ=+0.114). This empirically demonstrates a fundamental principle of ensemble learning: **the base learners in a stacking ensemble should be diverse and individually strong**. RF was the weakest model and contributed noisy, unreliable predictions as meta-features — diluting the ensemble rather than enriching it. The Ridge meta-learner also selected a near-zero alpha (0.001) for Stack V2 vs. a stronger regularisation (1.0) for Stack V1, suggesting V2's meta-features were cleaner and less noisy.[ppl-ai-file-upload.s3.amazonaws]

ANN's Competitive Performance is Noteworthy

The ANN achieved R^2=0.9401 for height and R^2=0.8039 for aspect ratio — competitive with the best-performing GPR. The use of **early stopping with a held-out validation fraction** (15%) was essential here; without it, the ANN would overfit at N=89. The architecture selected by GridSearchCV — (128, 64) neurons, ReLU activation, α=0.0001 — is a relatively deep configuration for this dataset size. For the thesis, this demonstrates that with proper regularisation, neural networks remain viable even on small datasets, but require careful hyperparameter control.[ppl-ai-file-upload.s3.amazonaws]

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5. Statistical Significance Analysis (Wilcoxon Signed-Rank)

The Wilcoxon signed-rank test was applied pairwise across all models to determine whether performance differences are statistically meaningful, rather than attributable to chance.[ppl-ai-file-upload.s3.amazonaws]

Key findings for Height ($p < 0.05$ = significant):

- GPR vs SVR: p=0.0001 → GPR is significantly better
- GPR vs RF: p=0.0000 → GPR is significantly better
- GPR vs ANN: p=0.0065 → GPR is significantly better
- GPR vs Stack V2: p=0.0139 → GPR is significantly better

- SVR vs RF: p=0.0915 ✗ → Not significantly different — both are poor

Key findings for Aspect Ratio:

- GPR vs ANN: p=0.6162 ✗ → **Not significantly different** — critical finding
- GPR vs SVR: p=0.0014 → GPR is significantly better
- SVR vs Stack V2: p=0.7824 ✗ → Not significantly different despite apparent metric differences

The most important statistical finding is that **GPR and ANN are statistically indistinguishable for aspect ratio** (p=0.6162). This means that while GPR appears numerically superior (MAE 0.0379 vs 0.0416), this difference is not statistically robust given N=89. For the thesis, you should present GPR as the preferred model based on parsimony and theoretical fitness, but acknowledge that ANN is a statistically equivalent alternative for aspect ratio prediction.[ppl-ai-file-upload.s3.amazonaws]

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6. Improvement Analysis

Improvement B: Learning Curves — Data Sufficiency Diagnosis

Learning curves measure how model performance evolves as training set size increases. The notebook tracks 10 points from ~13 to 89 samples and diagnoses the trend. This is one of the most practically important analyses for a Master's thesis, as it directly addresses the question: **"should I collect more experimental data?"**[ppl-ai-file-upload.s3.amazonaws]

The diagnosis logic applied:

- If CV R^2 is **still rising** at N=89 → more data would help
- If CV R^2 has **plateaued** → current N is sufficient
- If **train–test gap is large** → overfitting, consider regularisation

For GPR, the expected outcome is a converged learning curve with a small gap, because GPR's kernel-based prior efficiently generalises from few samples. Document the specific diagnosis message from the plot in your thesis — it provides a principled, data-driven justification for the experimental budget.

Improvement C: SHAP Feature Importance (Bug-Fixed)

The critical bug in the previous version was a **feature name mismatch** — 3 names were provided for a 4-column feature matrix (the engineering feature `Heat_Input` was unnamed). This would have caused SHAP to crash or produce misaligned importance plots. The fix confirms `X_scaled.shape[1] == len(FEATURE_NAMES) == 4` before running any SHAP computation.[ppl-ai-file-upload.s3.amazonaws]

Two explainer types were correctly used:

- **TreeExplainer** for RF: exact, computationally efficient (uses tree structure directly)

- **KernelExplainer + k-means background** for ANN: model-agnostic, slower but correct for neural networks; k-means compresses 89 samples to 30 representative background points[ppl-ai-file-upload.s3.amazonaws]

The SHAP plots will reveal whether `Heat_Input (P/S)` — the physics-derived feature — carries disproportionate importance. If it ranks as the top feature, this confirms that **energy density is the primary physics driver**, validating the feature engineering choice and providing a strong narrative thread for the thesis.

Improvement D: Multi-Output GPR — The Most Significant Finding

Target	Multi-GPR R^2	Single-GPR R^2	Δ R^2	Multi-GPR MAE	Single-GPR MAE	Δ MAE
Height (µm)	**0.9682**	0.9552	+0.013	**58.83**	64.63	**−5.80 µm**
Aspect Ratio	**0.9689**	0.8739	+0.095	**0.0283**	0.0379	**−0.0096**

The Multi-Output GPR improves **both** targets simultaneously. The aspect ratio improvement of +0.095 R^2 is particularly significant — a 10.9% increase in explained variance. This finding has a direct physical interpretation: **bead height and aspect ratio are not independent response variables**. They share underlying physics (melt pool dynamics, solidification behaviour, heat conduction) that manifests as cross-target correlation. The joint GPR model implicitly captures this correlation structure through its shared kernel representation.[ppl-ai-file-upload.s3.amazonaws]

For the thesis, this is a publishable-quality insight: it suggests that **future optimisation should treat height and aspect ratio as coupled objectives**, not independent scalar outputs. This motivates the multi-objective optimisation (NSGA-II/PSO) planned for the next phase.

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7. Residual Analysis & Normality

All models produced **non-normally distributed residuals** (Shapiro-Wilk $p << 0.05$ for all). This has several implications:[ppl-ai-file-upload.s3.amazonaws]

- **Standard confidence intervals** (which assume Gaussian residuals) are not valid for these models
- **GPR's built-in uncertainty bands** (posterior variance) are a better uncertainty quantification tool than parametric CIs — this is another advantage of GPR in this context
- The non-normality may reflect **heteroscedasticity**: prediction error may be larger at extreme parameter combinations (high power + high speed), where the process physics becomes more complex
- For the thesis, this should be flagged as a limitation, and prediction intervals should be obtained from **GPR's posterior distribution** rather than from residual-based CIs

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8. Limitations & Risk Factors for the Thesis

- **N=89 is small** by ML standards. All performance estimates have wide confidence intervals. The 5-fold CV means each validation fold has only ~18 samples — insufficient for stable metric estimation. Consider reporting metric standard deviations alongside means.
- **No external validation set**: All evaluation is based on cross-validation within the same 89 samples. Reporting the model as validated without a held-out test set is a scientific risk. If possible, reserve 10–15 samples as a true holdout before any model training.
- **Residual non-normality**: As noted above, this invalidates standard parametric inference. Quantify uncertainty using GPR posterior bands or bootstrap resampling.
- **Aspect ratio performance is weaker**: The best single-output model achieves R^2=0.8739 — acceptable, but not excellent. The aspect ratio is a ratio of two measured quantities (width/height or related), making it more susceptible to measurement noise propagation. Consider whether the raw measurements (width, height) are better modelled separately.
- **Heat Input as a derived feature**: While physically motivated, P/S introduces a near-collinearity with Power and Speed individually. Regularised models (GPR, Ridge meta-learner) handle this well, but RF's variable importance may be distorted by collinearity.

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9. Next Steps & Thesis Progression

The notebook outputs the best models as serialised `.pkl` files, ready for deployment in the next phase. The recommended progression is:[ppl-ai-file-upload.s3.amazonaws]

1. **Inverse optimisation**: Use the saved GPR and Multi-Output GPR as surrogate models inside a **NSGA-II or PSO multi-objective optimiser** to identify Pareto-optimal process parameter combinations that simultaneously minimise height error and aspect ratio deviation from target
2. **Uncertainty-aware optimisation**: Exploit GPR's posterior variance to implement **Expected Improvement (EI)** acquisition functions — turning the process optimisation into a Bayesian Optimisation problem
3. **Experimental validation**: Select 5–10 Pareto-optimal parameter sets, run physical LMD experiments, and compare predicted vs actual bead geometry — this constitutes the validation chapter
4. **Sensitivity analysis**: Use the SHAP outputs to inform a **process window** analysis: at what parameter combinations does the model predict the most reliable (low uncertainty) outcomes?
5. **Consider extending N**: If the learning curve analysis shows CV R^2 still rising at N=89, additional targeted experiments in under-explored regions of the parameter space (active learning strategy) would directly improve model reliability